THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING

DEPARTMENT OF CIVIL ENGINEERING

CIVL1113: ENGINEERING MECHANICS AND MATERIALS

Time: December 16, 2020 9:30 AM – December 17, 2020 9:30 AM (24 hours)

Answer any **5 (FIVE)** questions.
All questions carry equal marks
Marks for sub-questions (if any) are given.

Note: The answer must be handwritten, including the figures. Please submit one word or pdf document. Compiling photos in one word document is allowed.

Students should independently finish the take-home exam.

For the mechanics part, detailed reasoning, equations and steps are necessary.

SECTION A

Q1 As case 1 shown in Figure 1(a), two beams are piled up without bonding in the interface, see Figure 1(b) for the cross sections, whereas for case 2 in Figure 1(c), they are pinned together with the cross sections shown in Figure 1(d). For each case, the two beams are fixed to the wall at point A, and a concentrated load F is applied at the free end B. The upper beam and bottom beam have the same Young's Modulus E but different thicknesses, $4h$ and h respectively. Young's Modulus E=200 GPa, width of cross section b=6 cm, h=2 cm, beam span L=1m.

(a) If F=1.008 kN, calculate the deflection $w_{B,1}$ at point B for case 1. **(4 marks)**

(b) Using the same load F in Question (a), calculate the deflections $w_{B,2}$ at point B for case 2. **(4 marks)**

(c) If the allowable stress for the material is $[\sigma]$=250 MPa, find the maximum allowable load $F_{1,max}$ and determine the failure surface for case 1. Only consider strength condition and assume the tensile strength equals to the compressive strength. **(5 marks)**

(d) Using the same allowable stress $[\sigma]$ in Question (c), find the maximum allowable load $F_{2,max}$ and determine the failure surface for case 2. Only consider strength condition and assume the tensile strength equals to the compressive strength. **(5 marks)**

(e) What can be concluded after the analysis comparing case 1 and 2? **(2 marks)**

Q2 A vertical bar AB is pin connected with a long horizontal beam. The other end of the vertical bar is fixed, as shown in Figure 2(a). The cross section of the solid vertical bar is rectangular as shown in Figure 2(b). The length of the vertical bar is 3 meters. The modulus E is 13 GPa and the yield strength is 30 MPa.

(a) Determine the maximum compressive loading the vertical bar could resist. **(15 marks)**

(b) Determine the failure mode of the vertical bar. **(5 marks)**

Q3 A bar is fixed at one end. At the other end, it is subjected to both Torsion $\boldsymbol{T}$ and bending moment $\boldsymbol{M}$, as shown in Figure 3. The diameter of the bar is $\boldsymbol{D}$.

(a) Determine the positions where the maximum torsional shear stress is and the values. **(5 marks)**

(b) Determine the positions where the maximum bending stress is and the values. **(5 marks)**

(c) Determine the positions where the maximum principle stress is and the values. **(5 marks)**

(d) Determine the positions where the minimum principle stress is and the values. **(5 marks)**

FIGURES

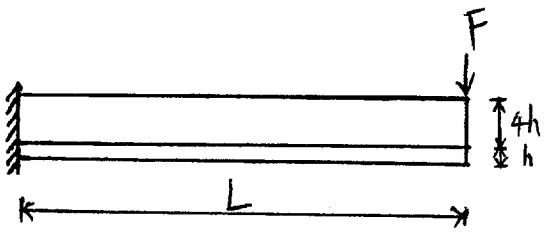


Figure 1(a) Case 1

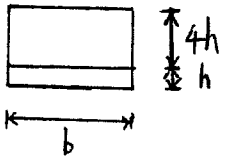


Figure 1(b) Cross section of case 1

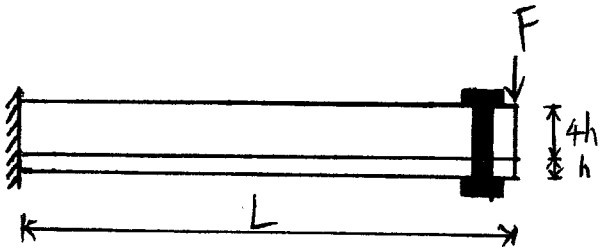


Figure 1(c) Case 2

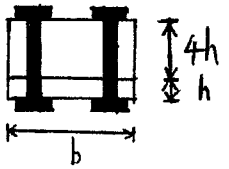


Figure 1(d) Cross section of case 2

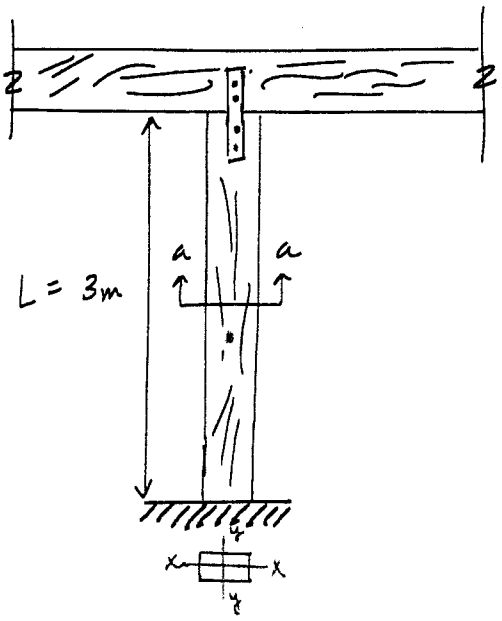


Figure 2(a)

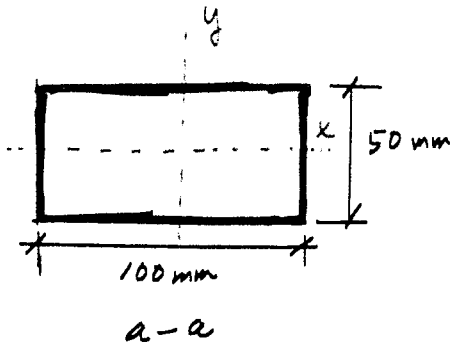


Figure 2(b)

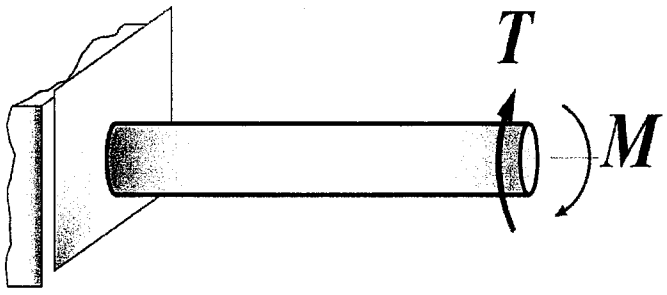


Figure 3

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